

Marcus Cheng

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Electrical Engineering

Dean's Honour List | Scholarships Recipient | Teaching Assistant | Design Team Lead | Undergrad Researcher

Vancouver, BC

Sep. 2021 – April 2027

WORK EXPERIENCE

Undergrad Power Electronics Researcher

Power Electronic Systems and Informatics Laboratory (PESIL)

May 2025 – Aug 2025

Vancouver, BC

- Researched and designed next-gen gate driver circuits to reduce cost and size of solid-state DC circuit breakers
- Evaluated trade-offs across multiple gate-driver architectures via literature review, guiding component selection and design strategy
- Modeled gate-driver interactions in LTspice, capturing parasitic effects, power-transistor dynamics, and parallel varistor behavior to ensure reliable operation under transient conditions

Package Engineering Intern (SI/PI & Layout)

Tenstorrent

Jan. 2024 – Aug. 2024

Toronto, ON

- Designed die-to-die (D2D) interconnect and PDN layouts for advanced and standard packages using Cadence Allegro Advanced Package Designer (APD), by considering SI/PI constraints and design rules such as trace-to-trace overlap, length matching, and maximum allowed via stack
- Studied the effect of variables such as signal to ground ratio, RX/TX termination, channel length, and data rate, on eye height & width for a 2.5D chiplet die-to-die interconnect by performing parameter sweeps using Ansys electronics desktop (AEDT) HFSS
- Developed guidelines, tips, and design flows to accelerate PDN layout and APD adoption across the team

Teaching Assistant

UBC CPEN 211 - Introduction to Microcomputers

Sep. 2023 – Dec. 2023

Vancouver, BC

- Supervised laboratory sessions and hosted office hours to help over 100 students to learn the basics of digital systems and microcomputers including breadboard circuit design, combinational and sequential circuits, SystemVerilog

PROJECTS & DESIGN TEAM

Dynamic Channel Emulator | *Link Budget, VNA, SDR, Channel Modeling*

Sep. 2025 – Apr. 2026

- Developed a Dynamic Channel Emulator (DCE) to simulate satellite-to-ground communication effects (e.g. Doppler shift, fading, path loss, delay) for testing LoRa-based IoT hardware under realistic Earth-space conditions

Thrust Vectoring Rocket Project Manager

UBC Rocket

Sep. 2025 – Mar. 2026

Vancouver, BC

- Co-led a team of 33 to develop a self-landing drone rocket powered by dual counter-rotating electric motors, serving as a testbed for advancing toward a self-landing liquid-fueled rocket

Avionics Hardware Lead

UBC Rocket

Sep. 2024 – Mar. 2026

Vancouver, BC

- Co-led a team of 25 to build the avionics hardware for Cloudburst, our rocket for the year of 2025/2026
- Designed and prototyped a next-generation live-telemetry and video system with >45,000 ft range, which involve link budgeting and speccing out RF parts such as LNAs and helical antennas given extreme space constraint of a 4-inch diameter rocket.
- Designed, assembled, and validated the next generation 8-layer, 2-sided flight controller PCB that is 30% smaller yet supports double the sensor count, delivering higher redundancy and more efficient use of space
- Designed and validated the inaugural battery management system for the rocket's avionics, which featured over-current and under-voltage protection

Self-Balancing Robot | *PID tuning, Kalman Filter, IMU calibration*

Mar. 2025 – Apr. 2025

- Built a robot with real-time PID control, increasing IMU sampling by 1600% and achieving $\pm 2^\circ$ stability.
- Implemented a Kalman filter for sensor fusion to enhance control accuracy.

TECHNICAL SKILLS

Hardware: AC/DC, Transient Analysis, PCB Schematic & Layout & Validation/Bring-up, SMT and THT Soldering, Oscilloscope, Signal Generator, Vector Network Analyzer, Multimeter, PC assembly and troubleshooting, I^2C , SPI , $UART$, SystemVerilog

Software: Altium Designer, Ansys HFSS, Cadence APD & ADE, LTspice, PSIM, Markdown, Git, GitHub, Python, MATLAB, Linux

Certifications: CompTIA A+ & Network+, Ham Radio License, Drone Pilot Certificate, Boat License